COLLEGE OF ENGINEERING
DEPARTMENT

CMPS 360 - Data Science Fundamentals

Fall 2026

Instructor Information

Name: Dr. Abdelkarim Erradi
Academic Title: Associate Professor
Office: H07 - C309, College of Engineering Building
Phone: 4403-4254
E-mail: erradi@qu.edu.qa
Office Hours (to be confirmed):

- L01 - Sunday 1:30pm to 2:30pm at Male office hours area
- L51 - Tuesday 1:30pm to 2:30pm at H07-C309

TA Information

Name: TBD
Office:
Phone:
E-mail:
Office Hours:

Class/Laboratory Schedule

L51 UTR 03:00 PM - 03:50 PM at H07- College of Engineering A140
L01 UTR 04:00 PM - 04:50 PM at H07- College of Engineering BL024

Coordinator Information

Same as the instructor

Course Information

Catalog Description:
Fundamental principles, algorithms, and methods of data science. Topics include the data science lifecycle, data acquisition and representativeness, exploratory analysis, visualization, sampling, statistical inference, hypothesis testing, and causal reasoning. Core analytical techniques such as regression, classification, and predictive modeling are introduced. Data ethics, including privacy, fairness, and responsible data use, are emphasized throughout. Hands-on projects using real-world datasets develop practical skills in analyzing, visualizing, and communicating data-driven insights with modern data science tools.

Credits: 3 credit hours.

Contact Hours: (3 hour theory, 0 hour lab).

Prerequisites: CMPS 151 - Programming Concepts

Textbook(s):
Ani Adhikari and John DeNero, *Computational and Inferential Thinking: The Foundations of Data Science*. Available at https://www.inferentialthinking.com/

References:
- Lau, S., Gonzalez, J., & Nolan, D. (2023). *Learning Data Science: Data Wrangling, Exploration, Visualization, and Modeling with Python*. O'Reilly Media. ISBN: 9781098113001. Available at: https://learningds.org/intro.html
- Igual, L., & Seguí, S. (2024). *Introduction to Data Science: A Python Approach to Concepts, Techniques and Applications* (2nd ed.). Springer Nature. ISBN: 978-3-031-48955-6. Available at: https://link.springer.com/book/10.1007/978-3-031-48956-3
- Trevor Hastie, Robert Tibshirani, Jerome Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd Edition, Springer, 2009. ISBN: 978-0387848570. Available online here.

Course Objectives:
- Apply computational, statistical, and data science techniques to acquire, prepare, analyze, visualize, and interpret real-world data using responsible and modern practices.
- Design, implement, evaluate, and communicate data science solutions using appropriate tools, methods, and workflows to address real-world problems.

Course Learning Outcomes (CLOs):
1. Explain fundamental data science concepts, including the data science lifecycle, data quality, exploratory analysis, statistical inference, modeling, and responsible data use.
2. Apply computational, statistical, and data science techniques to acquire, prepare, analyze, visualize, and interpret real-world data.
3. Design and implement data science workflows to address data-driven problems and evaluate and communicate their results effectively.

Relationship of Course Learning Outcomes (CLOs) to Student Outcomes (SOs):

Course Learning Outcomes (CLO)	Related CS Student Outcomes (SO)						Related AI Student Outcomes (SO)					
	1	2	3	4	5	6	1	2	3	4	5	6
1	√			√			√			√		
2		√						√				
3			√		√	√			√		√	√

BSc in CS Student Outcomes (CS-SOs)

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

BSc in AI Student Outcomes (AI-SOs)

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply artificial intelligence theories, models, and techniques to design, implement and evaluate AI-based solutions that solve complex problems.

<u>Topics Covered:</u>

Topics	**Chapter***	**Section***	**Weeks**
Data Science Foundations: Applications, Roles, Lifecycle, and Reproducible Workflows			1
Data Problem Formulation, Acquisition, Quality, and Representativeness			1
Data Wrangling: Cleaning, Transformation, Integration, and Feature Engineering			1
Exploratory Data Analysis and Visualization			2
Probability, Sampling, Uncertainty, and Statistical Inference			1.5
Statistical Testing, Estimation, and A/B Testing			1
Introductory Predictive Analytics: Regression and Classification			1.5
Pattern Discovery: Clustering and Dimensionality Reduction			1
Causal Inference and Experimental Design			2
Working with Non-Tabular Data such as Text, and Images			1.5
Responsible Data Science: Privacy, Fairness, Transparency, and Communication			0.5

Total			**14**

*Optional

Method of Instruction

The course uses a **hands-on, application-oriented approach** combining lectures, demonstrations, practical assignments, and problem-based learning. Students apply computational, statistical, and data science techniques to real-world datasets through Python-based data analysis, visualization, statistical modeling, causal analysis, and non-tabular data applications. An **integrated data science project** reinforces problem formulation, critical thinking, responsible data practices, and effective communication of results.

Assessment Methods and Grading Policy

Assignments:	15% - 3 assignments
Quizzes:	15% - 4 take 3 (no makeup for missed quizzes)
Project:	15% - Report and Presentation
Midterm Exam:	20% - Week 7
Final Exam:	35%

ABET Contribution of Course to Professional Component

Subject Area (Credit Hours)

Math & Basic Science	: 10%
Engineering	: 10%
Computer Science	: 80%
Engineering Design	: 0%
Broad Education	: 0%

Computer/Software Usage

Python-based tools for data analysis, visualization, and modeling, including Jupyter Notebook or Google Colab and core libraries such as NumPy, Pandas, Matplotlib/Seaborn, and scikit-learn.
Course materials, submissions, and announcements will be managed through Blackboard, GitHub and Microsoft Teams.

Laboratory Projects

N/A.

Course Ground Rules

- Students are expected to participate actively, ask questions, and engage respectfully during lectures and discussions.
- Students should come to class prepared by reviewing assigned materials, completing required readings or pre-class tasks, and bringing the tools needed for Python-based work.

- **Attendance:** Attendance is mandatory according to University policies. Students who exceed the permitted absence limit may not qualify for course credit.
  Late arrival after attendance has been taken may be counted as an absence for that day.
- **Communication:** Students must check Blackboard and Teams regularly for announcements, materials, deadlines, and course updates.
- **Submissions:** Assignments and project deliverables must be submitted on time, in the required format, and through the designated platform.
- **Data and Code Practice:** Students must write clear, reproducible, and well-documented code; cite data sources; explain assumptions; and submit work that can be reviewed and executed when applicable.
- **Academic Integrity:** Copying code, notebooks, reports, visualizations, or written answers from classmates, websites, repositories, or AI tools without authorization and acknowledgment is considered academic misconduct.
- **Generative AI Use Policy:** Generative AI may be used as a learning aid, **but not to replace students' independent work, reasoning, analysis, or original writing**.
  - **Allowed:** Use AI to clarify concepts, review syntax, brainstorm approaches, improve writing clarity, and debug or explain code.
  - **Not allowed:** Do not use AI to generate complete solutions, perform the main analysis, or produce assignment, quiz, exam, notebook, report, visualization, or project work for submission. If AI performs the analysis or solves the problem for you, you are not developing the skills this course is designed to teach.

University Code of Conduct

QU expects its students to adopt and abide by the highest standards of conduct in their interaction with professors, peers, staff members and the wider university community. Moreover, QU expects its students to act maturely and responsibly in their relationships with others. Every student is expected to assume the obligations and responsibilities required from them for being members of the QU community.

As such, a student is expected not to engage in behaviors that compromise their integrity, as well as the integrity of QU. Further information regarding the University Code of Conduct may be found on the web here.

Use of Artificial Intelligence (AI) Tools

In accordance with Article 6 of the Student Code of Conduct at Qatar University, academic violations include a range of actions, one of which pertains to submitting work that is not the individual's own production. This includes using creative artificial intelligence tools such as ChatGPT to produce content, images, videos, or programming code and presenting it as original work.

Therefore, students are cautioned that using artificial intelligence tools such as ChatGPT or any similar tools to produce academic content and present it as their own work is considered plagiarism, exposing the student to disciplinary penalties as stipulated in Qatar University's Student Code of Conduct. In light of this, we urge all students to adhere to ethical standards in all assignments and academic work, and to seek guidance from the course instructor when unsure about the proper and ethical use of artificial intelligence sources in completing assignments, duties, and academic tasks. Further information regarding Instructions on the use of Generative Artificial Intelligence Tools at Qatar University may be found on the web here.

Support for Students with Special Needs

It is Qatar University policy to provide educational opportunities that ensure fair, appropriate and reasonable accommodation to students who have disabilities that may affect their ability to participate in course activities or meet course requirements. Students with disabilities are encouraged to contact their Instructor to ensure that their individual needs are met. The University through its Inclusion and Special Needs Support Center will exert all efforts to accommodate for individuals' needs.

Contact Information for Inclusion and Special Needs Support Center:
Tel-Female: (+974) 4403 7972 – 4403 3854
Tel-Male: (+974) 4403 7946 – 4403 7942
Location: Student Affairs Building (I11) – Third Floor
Email: specialneeds@qu.edu.qa

Academic Support and Learning Resources

The University Student Learning Support Center (SLSC) provides academic support services to male and female students at QU. The SLSC is a supportive environment where students can seek assistance with academic coursework, writing assignments, transitioning to college academic life, and other academic issues. SLSC services include Peer Tutoring, the Writing and Language support, Writing Workshops, and Academic Success Workshops. Students may also seek confidential Academic Coaching from the professional staff at the Center.
Contact Information for Students Support and Learning Resources:
Tel-Female: (00974) 4403 3870
Tel-Male: (00974) 4403 7963
Location: Student Affairs Building (I11) – First Floor
E-mail: learningcenter@qu.edu.qa

College of Engineering Learning Support

The Engineering Success Oasis (ESO) provides academic support services to all students registered in Engineering courses at QU. We provide academic assistance through group and one-on-one tutoring, tailored major programs, and various workshops. Support schedules are announced at the beginning of semesters.

Contact Information for College of Engineering Learning Support:
Females
Tel: (+974) 4403 6380
Email: CENG.SuccessOasis.F@qu.edu.qa
Location: H07 - C242
Support Hours: 9:00 am to 4:00 pm

Males
Tel: (+974) 4403 6389

Email: CENG.SuccessOasis.M@qu.edu.qa
Location: H07 - B114
Support Hours: 9:00 am to 4:00 pm

Sessions' Booking
Females
1- One-to-one sessions' registration:
https://n9.cl/female_ceng_success_oasis
2- Weekly sessions via email invitation from Engineering Success Oasis (ESO)

Males
1- One-to-one sessions' registration:
https://n9.cl/male_ceng_success_oasis
2- Weekly sessions via email invitation from Engineering Success Oasis (ESO)

Student Complaints Policy

Students at Qatar University have the right to pursue complaints related to faculty, staff, and other students. The nature of the complaints may be either academic or non-academic. For more information about the policy and processes related to this policy, you may refer to the student handbook.

Declaration

This syllabus and contents are subject to changes in the event of extenuating circumstances. The instructor (with approval of the Head of Department) reserves the right to make changes as necessary. If changes are necessitated during the term of the course, the students will be notified by email communication and posting the notification on the online teaching tool Blackboard. It is the student's responsibility to check on announcements made while they were absent.

Faculty Name: Dr. Abdelkarim Erradi
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